INJECTOR TUBES

To insure efficient cooling, each fuel injector is seated in a thin walled copper tube which extends through the water space in cylinder head as shown in figure 71. The lower end of the copper tube is pressed into cylinder head, and spun over. The upper end is flanged and sealed with a Neoprene seal. The spun-over lower end, and sealed upper end, prevents water leaks around the copper tube.

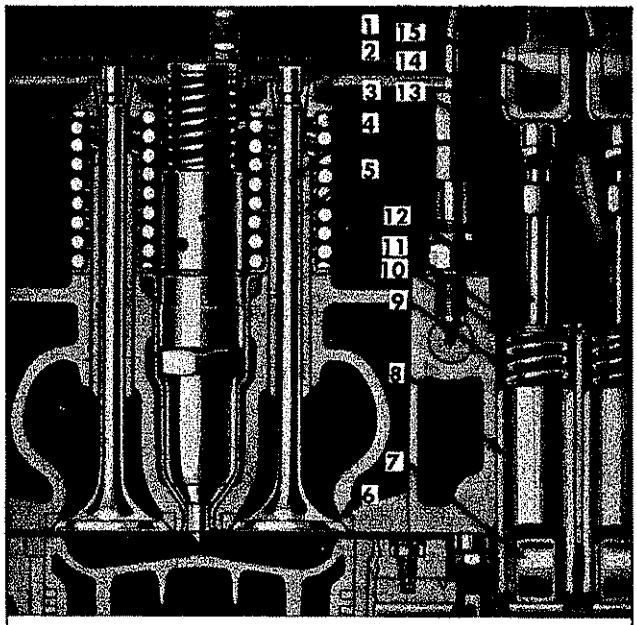
WATER NOZZLES

In addition to being surrounded by cooling liquid, cooling of injector tubes, valve seats, and valve guides is further assured by the use of water nozzles in cylinder head (fig. 71). Nozzles are positioned in such a manner that cool water entering the cylinder head is directed against these sections which are subject to greatest heat.

CYLINDER HEAD GASKETS

Combustion in each cylinder is individually sealed by Belleville cupped type ring seals, which are installed on top of cylinder liner and seals compression. Synthetic rubber rings and seals are installed in counterbores in cylinder block and effectively seal oil and water passages between cylinder block and cylinder head. An additional gasket around outer edge of head seals against leakage of oil (fig. 72).

The top of each cylinder head is completely enclosed by a removable pressed steel valve rocker cover. Cover is held in place by screws fitted with hand knobs, and is sealed against leakage by a gasket, held in place by a retainer and flanged edge of the cover.



1 Spring Lock	9 Push Rod Spring
2 Spring Cap	10 Spring Seat
3 Valve Spring	11 Retainer
4 Exhaust Valve	12 Push Rod
5 Valve Guide	13 Lock Nut
6 Valve Insert	14 Push Rod Clevis
7 Follower Roller	15 Valve Rocker Arm
8 Cam Follower	

Figure 70—Exhaust Valves, Guides and Inserts in Cylinder Head

REMOVAL

Equipment and accessories on the engine varies with individual installations to such an extent that no attempt is made to specifically list the steps necessary to remove the head. However, the following general procedures may be used.

1. Remove governor control housing assembly as directed under "Removal" in respective governor sections in **FUEL SYSTEM** section of this manual.
2. Remove cap screws attaching cylinder head lifting brackets to flywheel and balance weight covers.

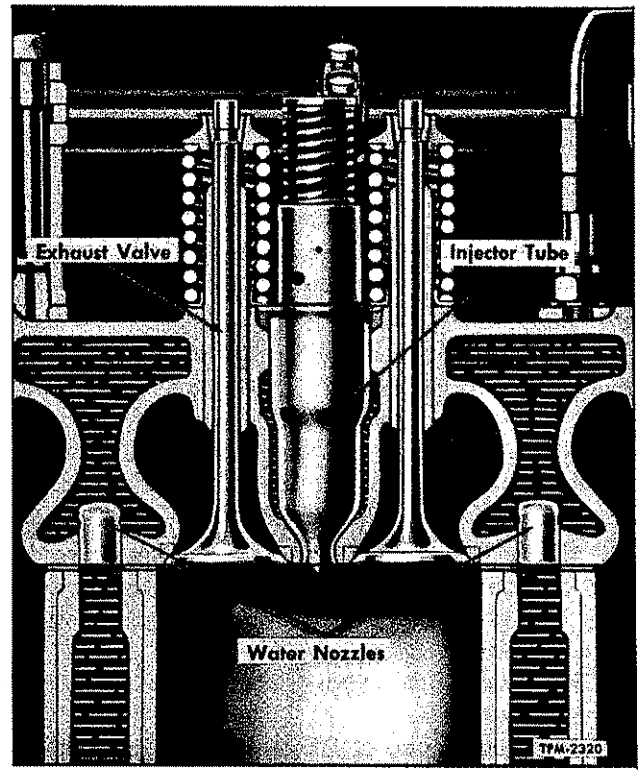


Figure 71—Water Cooling at Valves and Injector in Cylinder Head